

Combinatorial Optimization

Graphs

Paths

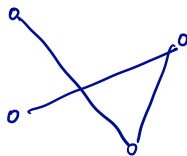
Cuts

Subsets

Assignments

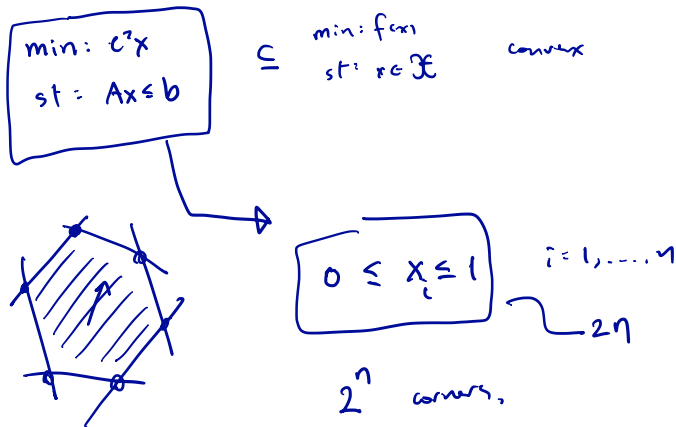
Schedules

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This course: Algorithms, Geometry and Polyhedral Techniques

The Role of Linear Optimization



Combinatorial Optimization

Part I: Linear Optimization

Geometry

Duality

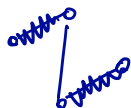
Algorithms

Part II: Polyhedral Combinatorics

Total Unimodularity, Total Dual Integrality

Bipartite and non-bipartite matching

Matroids and Matroid Polytope



Part III: Submodularity

Definitions, Extensions, Algorithms

Part IV: Extended Formulations

Part V: Randomized rounding and other topics



Main tools we need

Linear algebra

Optimization and duality

Mathematical maturity...



Requirements for the class

Homework: Every 2-3 weeks

Final Project: Expository

Due the last day of class

Useful References

- * Intro to Linear Opt.

- * Comb. Opt

- * Geom. Alg & Comb Opt

- * Comb. Opt.